

Sensors, Actuators and Embedded Platforms

1 Introduction

The Internet of Things (IoT) refers to a network of interconnected physical devices that can collect, exchange, process, and utilize data through communication networks. One of the most fundamental requirements of any IoT system is the ability to perceive and monitor the surrounding environment. This capability is provided by sensors.

Sensors act as the interface between the physical world and the digital world. They detect physical phenomena such as temperature, humidity, pressure, light intensity, motion, sound, vibration, and gas concentration, and convert these phenomena into electrical signals that can be processed by computing systems.

In modern IoT applications, sensors play a critical role in enabling intelligent decision-making. Smart homes, industrial automation systems, healthcare monitoring devices, environmental monitoring stations, and smart agriculture solutions all depend heavily on sensor-generated data.

Without sensors, IoT devices would have no means of acquiring information about their surroundings. Therefore, sensors are often regarded as the eyes and ears of an IoT system.

While sensors enable an Internet of Things (IoT) system to perceive its environment, actuators provide the capability to interact with and modify that environment. Sensors collect information such as temperature, humidity, pressure, motion, light intensity, and gas concentration, whereas actuators convert electrical signals into physical actions. Together, sensors and actuators form the foundation of cyber-physical systems in which digital computation influences real-world processes.

Consider a smart home system. A temperature sensor continuously monitors room temperature and sends data to a microcontroller. When the measured temperature exceeds a predefined threshold, the controller activates an air-conditioning unit through a relay. In this example, the temperature sensor serves as the sensing component, while the relay and air conditioner together perform the actuation function. Such closed-loop interaction between sensing and actuation is central to modern IoT systems.

Actuators are used extensively in industrial automation, robotics, smart agriculture, healthcare devices, autonomous vehicles, and smart buildings. Their operation often

involves the conversion of electrical energy into mechanical motion, thermal energy, hydraulic pressure, pneumatic force, or electromagnetic action.

2 Fundamentals of Sensors

2.1 Definition of a Sensor

A sensor is a device that detects, measures, and responds to changes in physical, chemical, or biological conditions and converts the observed quantity into a usable electrical signal.

The quantity being measured is known as the *measurand*. Examples of measurands include:

- Temperature
- Pressure
- Humidity
- Light intensity
- Gas concentration
- Sound level
- Motion
- Vibration

A sensor transforms information from the physical domain into a form that electronic systems can process and interpret.

2.2 Sensor as a Transducer

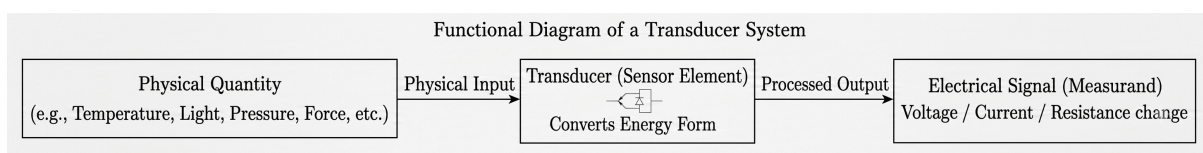


Figure 1: Sensor as a Transducer

A transducer is a device that converts one form of energy into another.

Sensors are a special class of transducers whose primary purpose is measurement.

Examples include:

- A microphone converts sound energy into electrical energy.
- A thermocouple converts thermal energy into electrical voltage.

- A photodiode converts light energy into electrical current.

Although the terms sensor and transducer are often used interchangeably, every sensor is a transducer, but not every transducer is necessarily a sensor. Figure 1 illustrates how a sensor acts as a transducer by converting a physical quantity into an electrical signal that can be processed by electronic systems.

2.3 General Working Principle of a Sensor

The operation of a sensor generally involves three stages:

1. Detection of a physical phenomenon.
2. Conversion of the phenomenon into an electrical signal.
3. Transmission of the signal to a processing unit.

The general flow of information may be represented as:

Physical Quantity → Sensor → Electrical Signal → Processing Unit

The processing unit may be a microcontroller, embedded system, computer, or cloud platform.

3 Role of Sensors in IoT Systems

Sensors form the perception layer of an IoT architecture. Their primary function is to collect information from the environment and provide it to higher layers of the system for processing and decision-making.

Figure 2 illustrates the position of sensors within a typical IoT architecture. Sensors collect information from the physical environment and provide it to embedded devices for processing. The processed data are transmitted through communication networks to cloud platforms, where storage, analytics, and decision-making operations are performed. Finally, the information is presented to end users through applications and dashboards. The information flow in a typical IoT system is shown below:

Sensor → Embedded Device → Network → Cloud Platform → Application

The collected data may be used for:

- Monitoring environmental conditions.
- Detecting abnormal events.
- Supporting automated decision-making.
- Controlling actuators.
- Predictive maintenance.
- Real-time analytics.

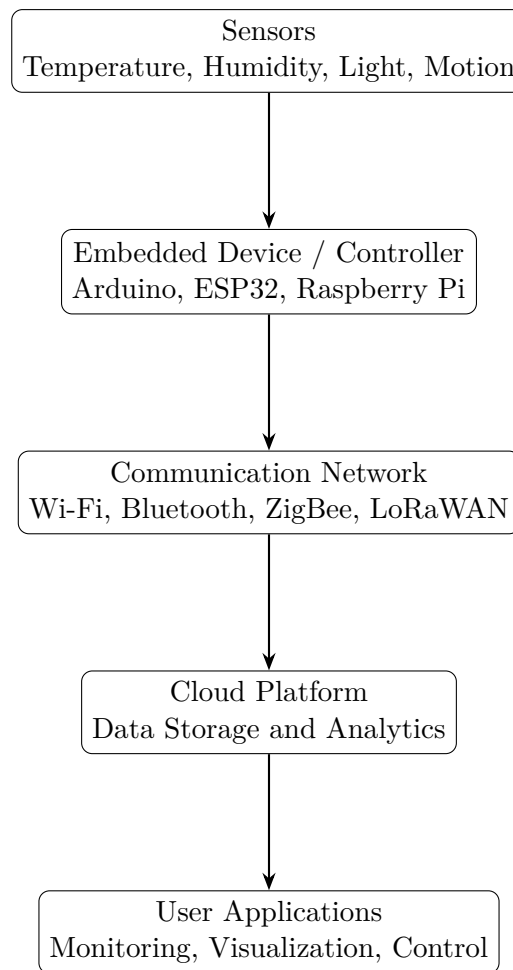


Figure 2: Role of Sensors in a Typical IoT Architecture

3.1 Examples of Sensor-Based IoT Systems

3.1.1 Smart Home

A smart home may use:

- Temperature sensors
- Motion sensors
- Light sensors
- Smoke sensors

to automatically regulate appliances and improve security.

3.1.2 Smart Agriculture

Agricultural IoT systems commonly employ:

- Soil moisture sensors

- Temperature sensors
- Humidity sensors
- Water level sensors

to optimize irrigation and crop productivity.

3.1.3 Industrial IoT

Industrial systems use sensors to monitor:

- Machine temperature
- Pressure
- Vibration
- Rotational speed

for predictive maintenance and fault detection.

4 Characteristics of Sensors

The performance of a sensor is evaluated using several important characteristics.

4.1 Accuracy

Accuracy indicates the closeness of a measured value to the true value of the quantity being measured.

For example, if the actual temperature is 30°C and a sensor reports 29.8°C, the measurement is considered highly accurate.

Higher accuracy generally leads to better measurement quality but often increases system cost.

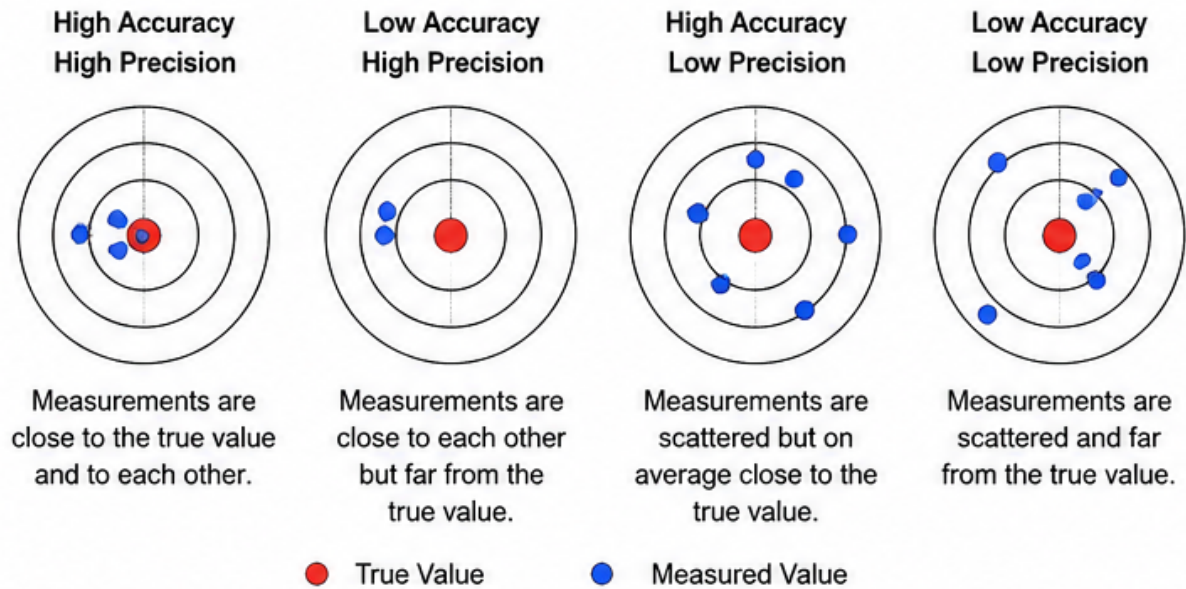
4.2 Precision

Precision refers to the ability of a sensor to produce consistent measurements under identical operating conditions. Consider the following repeated measurements:

28.5°C, 28.5°C, 28.5°C

These readings demonstrate high precision because they are identical.

Precision and accuracy are different concepts. A sensor may be highly precise but not necessarily accurate.



Note: Accuracy refers to closeness to the true value.
Precision refers to closeness of measurements to each other.

Figure 3: Accuracy Vs Precision

4.3 Sensitivity

Sensitivity represents the amount of change in sensor output produced by a corresponding change in input.

Mathematically,

$$S = \frac{\Delta O}{\Delta I}$$

where

- S denotes sensitivity,
- ΔO represents the change in output,
- ΔI represents the change in input.

A highly sensitive sensor can detect very small changes in the measured quantity.

4.4 Resolution

Resolution is the smallest detectable change in the measured quantity.

For example, if a digital temperature sensor displays readings in increments of 0.1°C , its resolution is 0.1°C .

4.5 Measurement Range

The measurement range specifies the minimum and maximum values that a sensor can measure reliably.

For example,

$$-40^{\circ}\text{C} \leq T \leq 125^{\circ}\text{C}$$

may represent the operating range of a temperature sensor.

4.6 Linearity

Linearity describes how closely the sensor output follows a straight-line relationship with respect to the input quantity.

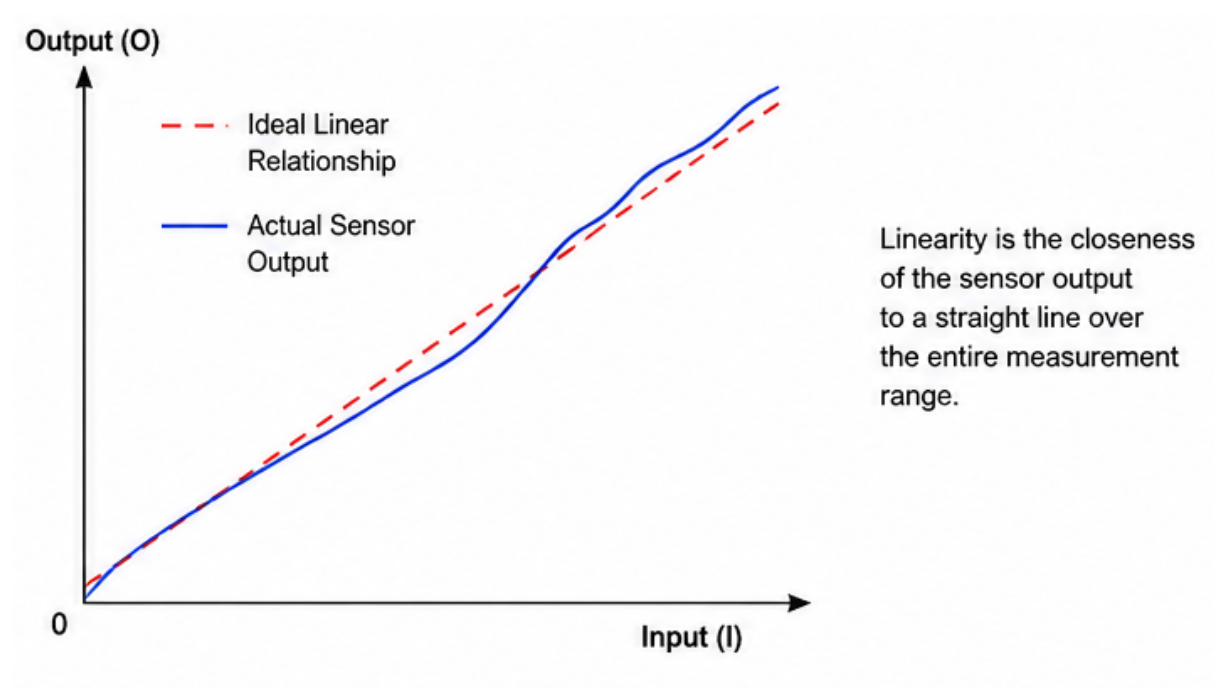


Figure 4: Linearity

Highly linear sensors simplify calibration and data interpretation.

4.7 Repeatability

Repeatability refers to the ability of a sensor to produce the same output under identical operating conditions over multiple measurements.

Good repeatability is essential for industrial monitoring and scientific measurements.

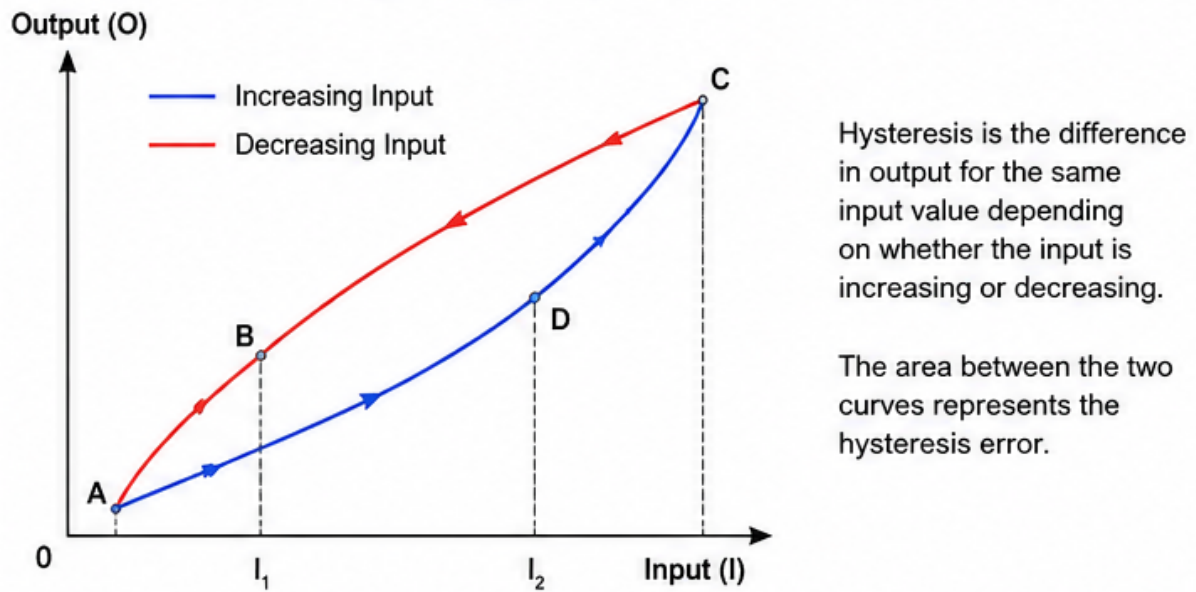


Figure 5: Hysteresis

4.8 Hysteresis

Hysteresis is the phenomenon in which the output of a sensor depends not only on its current input but also on its previous state.

This can result in different outputs for increasing and decreasing values of the same input quantity.

4.9 Drift

Drift refers to the gradual change in sensor output over time even when the input remains constant.

Drift may occur due to:

- Aging of components
- Environmental effects
- Mechanical wear
- Temperature variations

Periodic calibration is often required to compensate for drift.

4.10 Response Time

Response time is the time required by a sensor to respond to a change in the measured quantity and reach a specified percentage of its final value.

Fast response times are desirable in real-time monitoring and control systems.

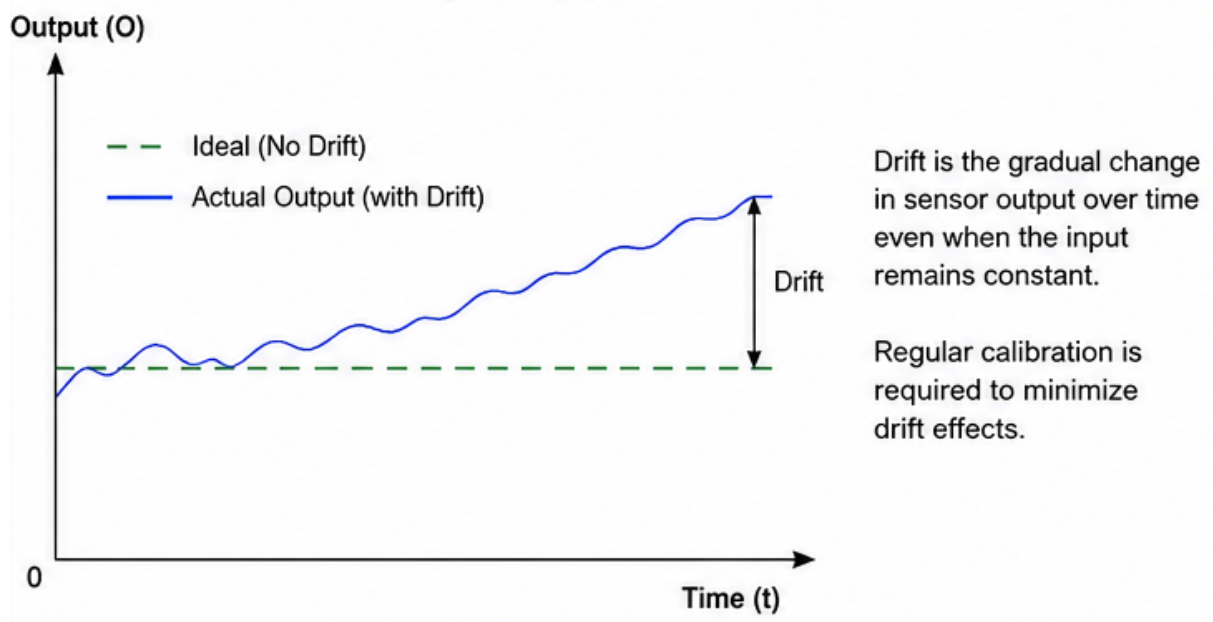


Figure 6: Drift

5 Classification of Sensors

Sensors can be classified in several ways.

5.1 Classification Based on Output Signal

5.1.1 Analog Sensors

Analog sensors generate continuously varying output signals.

Examples include:

- LM35 temperature sensor
- Light Dependent Resistor (LDR)
- Potentiometer

Advantages:

- Simple design
- Low cost
- Wide availability

Disadvantages:

- Susceptible to noise
- Requires analog-to-digital conversion

5.1.2 Digital Sensors

Digital sensors produce discrete digital outputs.

Examples include:

- DHT11
- DHT22
- DS18B20

Advantages:

- Better noise immunity
- Easier integration with digital systems
- Higher reliability in communication

5.2 Classification Based on Energy Source

5.2.1 Active Sensors

Active sensors generate output signals without requiring external excitation.

Examples:

- Thermocouple
- Piezoelectric sensor

5.2.2 Passive Sensors

Passive sensors require external power or excitation signals for operation.

Examples:

- LDR
- RTD
- Capacitive humidity sensor

5.3 Classification Based on Direction

5.3.1 Vector Sensors

A vector sensor measures physical quantities that possess both a magnitude and a specific direction in 2D or 3D space. To fully capture a vector quantity, these sensors typically use multiple sensing elements oriented along orthogonal axes (usually X , Y , and Z).

Examples:

- Accelerometer
- Magnetometer
- Wind Anemometer

5.3.2 Scalar Sensors

A scalar sensor measures physical quantities that can be completely described by a single numerical value (a magnitude) representing "how much" of something exists at a given point. These quantities do not have a spatial direction. These sensors A single real number.

Examples:

- Temperature Sensor
- Pressure Sensor
- Humidity sensor

5.4 Classification Based on Measured Quantity

Common categories include:

- Temperature sensors
- Humidity sensors
- Pressure sensors
- Light sensors
- Motion sensors
- Gas sensors
- Proximity sensors
- Sound sensors
- Biosensors

6 Sensor Selection Criteria

Selecting an appropriate sensor is an important aspect of IoT system design.

Several factors must be considered.

6.1 Measurement Requirements

The sensor should provide the required measurement range, resolution, and accuracy.

6.2 Power Consumption

Low-power sensors are preferred for battery-operated and remote IoT devices.

6.3 Environmental Conditions

Sensors must be capable of operating reliably under expected environmental conditions such as:

- High temperature
- High humidity
- Dust exposure
- Mechanical vibration

6.4 Cost

The overall cost of deployment becomes significant in large-scale IoT systems involving hundreds or thousands of sensors.

6.5 Communication Compatibility

The sensor should support suitable interfaces such as:

- GPIO (General Purpose Input/Output)
- UART (Universal Asynchronous Receiver Transmitter)
- SPI (Serial Peripheral Interface)
- I²C (Inter-Integrated Circuit)

6.6 Maintenance Requirements

Frequent maintenance and calibration increase operational costs and system complexity.

7 Challenges in Sensor-Based IoT Systems

Despite their usefulness, sensors present several practical challenges.

7.1 Noise and Interference

Electrical noise can distort sensor signals and reduce measurement accuracy.

Sources of noise include:

- Electromagnetic interference
- Power supply fluctuations
- Environmental disturbances

7.2 Calibration

Sensor outputs may deviate from true values over time. Periodic calibration is necessary to maintain measurement accuracy.

7.3 Power Constraints

Many IoT devices operate using batteries and therefore require energy-efficient sensing mechanisms.

7.4 Scalability

Large IoT deployments may generate enormous volumes of sensor data that require efficient storage, processing, and transmission.

7.5 Security Issues

Compromised sensors may provide incorrect information, potentially leading to incorrect decisions and system failures.

7.6 Environmental Effects

Extreme environmental conditions can degrade sensor performance and reduce operational lifetime.

8 Future Trends in Sensor Technology

Recent developments in sensor technology are driving the next generation of IoT systems.

Important trends include:

- MEMS-based sensors
- Smart sensors with embedded processing
- Ultra-low-power sensors

- Self-calibrating sensors
- AI-enabled sensing devices
- Flexible and wearable sensors
- Energy-harvesting sensor systems

These advancements are expected to improve sensing accuracy, reliability, intelligence, and energy efficiency.

9 Temperature Sensors

Temperature measurement is one of the most common sensing requirements in IoT systems. Applications include weather monitoring, industrial process control, smart agriculture, healthcare monitoring, and smart homes.

9.1 Working Principle

Temperature sensors exploit physical properties that vary with temperature, such as:

- Electrical resistance
- Semiconductor junction voltage
- Thermal expansion
- Infrared radiation

The measured variation is converted into an electrical signal that can be processed by a microcontroller.

9.2 Thermistors

A thermistor is a temperature-sensitive resistor whose resistance changes significantly with temperature.

9.2.1 Types of Thermistors

1. **Negative Temperature Coefficient (NTC)**

- Resistance decreases as temperature increases.

2. **Positive Temperature Coefficient (PTC)**

- Resistance increases as temperature increases.

9.2.2 Characteristics

- High sensitivity
- Low cost
- Non-linear response
- Suitable for moderate temperature ranges

9.2.3 Applications

- Smart thermostats
- Battery monitoring
- Home automation systems

9.3 LM35 Temperature Sensor

The LM35 is a precision integrated-circuit temperature sensor.

9.3.1 Features

- Linear output
- Output scale factor:

10 mV/°C

- No external calibration required

9.3.2 Working Principle

The internal semiconductor circuitry generates an output voltage proportional to temperature.

Table 1: LM35 Output Voltage

Temperature (°C)	Output Voltage (mV)
25	250
30	300
50	500

9.3.3 Advantages

- Simple interfacing
- Low power consumption
- Good accuracy

9.3.4 Applications

- Weather stations
- Industrial monitoring
- Smart appliances

9.4 Digital Temperature Sensors

Examples include:

- DS18B20
- TMP102
- MCP9808

9.4.1 Advantages

- Digital output
- Reduced noise susceptibility
- Easy interfacing using I²C, SPI, or One-Wire protocols

9.4.2 DS18B20 Temperature Sensor

Features:

- One-Wire communication
- Unique device address
- Multiple sensors on one bus
- Resolution up to 12 bits

Applications:

- Smart agriculture
- Data logging
- Industrial IoT

10 Humidity Sensors

Humidity sensors measure the amount of water vapor present in air.

10.1 Relative Humidity

Relative humidity (RH) is defined as:

$$RH = \frac{\text{Actual Water Vapor Content}}{\text{Maximum Possible Water Vapor Content}} \times 100\%$$

10.2 Capacitive Humidity Sensors

10.2.1 Working Principle

The dielectric constant of a moisture-absorbing material changes with humidity.

Since capacitance is given by

$$C = \frac{\epsilon A}{d}$$

changes in dielectric constant produce measurable capacitance variations.

10.2.2 Features

- High accuracy
- Long-term stability
- Wide operating range

10.3 DHT11 and DHT22 Sensors

These are popular low-cost digital temperature and humidity sensors.

10.3.1 DHT11

- Humidity range: 20–90
- Temperature range: 0 °C to 50 °C
- Low cost

10.3.2 DHT22

- Humidity range: 0–100
- Temperature range: −40 °C to 80 °C
- Higher accuracy

10.3.3 Applications

- Smart homes
- Agricultural monitoring
- HVAC systems

11 Light Sensors

Light sensors measure the intensity of visible or infrared radiation. They are widely used in smart lighting systems, environmental monitoring, industrial automation, and security applications.

11.1 Light Dependent Resistor (LDR)

An LDR, also known as a photoresistor, is a passive component whose resistance varies according to the intensity of incident light.

11.1.1 Construction

An LDR is typically made from semiconductor materials such as cadmium sulfide (CdS) or cadmium selenide (CdSe). The sensing surface is exposed to ambient light.

11.1.2 Working Principle

When photons strike the semiconductor material, additional charge carriers are generated, reducing the resistance of the device.

- High light intensity $\rightarrow$ Low resistance
- Low light intensity $\rightarrow$ High resistance

11.1.3 Characteristics

- Low cost
- Easy interfacing
- Wide availability
- Non-linear response
- Relatively slow response time

11.1.4 Applications

- Automatic street lighting
- Smart home illumination control
- Solar tracking systems
- Camera exposure control

11.2 Photodiodes

A photodiode is a semiconductor device that converts light energy directly into electrical current.

11.2.1 Working Principle

When photons strike the depletion region of a reverse-biased PN junction, electron-hole pairs are generated. These charge carriers produce a photocurrent proportional to the incident light intensity.

11.2.2 Characteristics

- Fast response
- High sensitivity
- Good linearity
- Suitable for optical communication

11.2.3 Applications

- Optical communication systems
- Light meters
- Medical instruments
- Industrial automation

11.3 Digital Light Sensors

Modern IoT systems frequently use digital light sensors that provide illumination measurements directly through digital interfaces.

Common examples include:

- BH1750

- TSL2561
- VEML7700

11.3.1 Advantages

- Direct digital output
- High measurement accuracy
- Reduced susceptibility to electrical noise
- Easy interfacing through I²C

11.3.2 Applications

- Smart buildings
- Adaptive display brightness control
- Energy management systems

12 Motion and Presence Sensors

Motion sensing is an essential requirement in modern IoT applications such as surveillance, occupancy detection, robotics, and smart home automation.

12.1 Passive Infrared (PIR) Sensor

A Passive Infrared (PIR) sensor detects changes in infrared radiation emitted by objects.

12.1.1 Working Principle

All objects with temperatures above absolute zero emit infrared radiation.

The PIR sensor contains pyroelectric materials that generate electrical signals when exposed to changing infrared energy levels. When a person moves across the sensor's field of view, the infrared radiation pattern changes, producing a detectable output signal.

12.1.2 Characteristics

- Low power consumption
- Wide detection area
- Low cost
- Easy integration with microcontrollers

12.1.3 Advantages

- Contactless detection
- Reliable human presence detection
- Long operational life

12.1.4 Limitations

- Cannot measure distance
- Performance affected by environmental conditions
- Limited ability to detect stationary objects

12.1.5 Applications

- Security systems
- Smart lighting
- Occupancy monitoring
- Automatic door control

13 Proximity and Distance Sensors

Proximity sensors detect nearby objects without physical contact, while distance sensors estimate the separation between the sensor and an object.

13.1 Ultrasonic Sensors

Ultrasonic sensors utilize high-frequency sound waves to measure distance.

A widely used example is the HC-SR04 ultrasonic sensor.

13.1.1 Working Principle

The sensor transmits an ultrasonic pulse and measures the time required for the reflected echo to return.

Distance is calculated using:

$$D = \frac{v \times t}{2}$$

where

D Distance to the object

v Speed of sound in air

t Round-trip travel time

The factor of two accounts for the forward and return paths.

13.1.2 Example

Assume

$t = 8 \text{ ms}$

and

$v = 343 \text{ m/s}$

Then

$$D = \frac{343 \times 8 \times 10^{-3}}{2}$$

$D = 1.372 \text{ m}$

13.1.3 Advantages

- Non-contact measurement
- Good accuracy
- Independent of object color
- Cost-effective

13.1.4 Applications

- Obstacle avoidance robots
- Water level monitoring
- Smart parking systems
- Industrial automation

13.2 Infrared Proximity Sensors

Infrared proximity sensors determine the presence of nearby objects using infrared radiation.

13.2.1 Working Principle

The sensor consists of:

- Infrared LED transmitter
- Photodiode or phototransistor receiver

The transmitter emits infrared light which is reflected by nearby objects. The receiver detects the reflected signal.

13.2.2 Advantages

- Compact design
- Fast response
- Low cost

13.2.3 Limitations

- Sensitive to surface properties
- Performance affected by ambient light
- Limited measurement range

13.2.4 Applications

- Object detection
- Robotic navigation
- Industrial conveyors
- Touchless switches

14 Pressure Sensors

Pressure sensors measure the force exerted per unit area.

Pressure is defined as:

$$P = \frac{F}{A}$$

where

P Pressure

F Applied force

A Area over which the force acts

14.1 Classification of Pressure Sensors

Pressure sensors may be classified as:

1. Piezoresistive sensors
2. Capacitive sensors
3. Piezoelectric sensors

14.2 Piezoresistive Pressure Sensors

These sensors exploit the piezoresistive effect in semiconductors.

14.2.1 Working Principle

Mechanical stress alters the resistance of the sensing element.

The resistance variation is converted into a measurable electrical signal using a Wheatstone bridge arrangement.

14.3 Capacitive Pressure Sensors

These sensors measure pressure-induced variations in capacitance.

14.3.1 Working Principle

Pressure deforms a diaphragm, changing the separation between capacitor plates and consequently altering capacitance.

14.4 Piezoelectric Pressure Sensors

These sensors generate electrical charges when subjected to mechanical stress.

14.4.1 Applications

- Dynamic pressure measurement
- Vibration monitoring
- Industrial diagnostics

14.5 BMP280 Pressure Sensor

The BMP280 is a popular digital environmental sensor.

14.5.1 Measured Parameters

- Atmospheric pressure
- Temperature

14.5.2 Communication Interfaces

- I²C
- SPI

14.5.3 Applications

- Weather monitoring stations
- Altitude estimation
- Drone navigation
- Environmental sensing

15 Gas Sensors

Gas sensors are used to detect the presence and concentration of gases in the surrounding environment. These sensors play a crucial role in environmental monitoring, industrial safety, smart cities, and healthcare applications.

15.1 MQ-Series Gas Sensors

The MQ-series is one of the most widely used families of gas sensors in IoT applications.

15.1.1 Working Principle

MQ sensors utilize a metal oxide semiconductor (MOS) sensing layer, typically tin dioxide (SnO₂).

When target gases interact with the sensing surface, the electrical resistance of the material changes. This variation is converted into a measurable electrical signal.

15.1.2 Characteristics

- High sensitivity
- Low cost
- Simple interfacing
- Requires calibration

- Moderate power consumption

15.1.3 Applications

- Air quality monitoring
- Industrial safety systems
- Smart homes
- Gas leakage detection
- Smart city deployments

16 Sound Sensors

Sound sensors convert acoustic energy into electrical signals that can be processed by embedded systems.

16.1 Microphone-Based Sound Sensors

Most IoT sound sensing modules use electret microphones or MEMS microphones.

16.1.1 Working Principle

Sound waves create pressure variations in air.

These pressure variations cause a diaphragm inside the microphone to vibrate. The vibrations are converted into electrical signals proportional to the sound intensity.

16.1.2 Characteristics

- Fast response
- Wide frequency range
- Compact size
- Low power consumption

16.1.3 Applications

- Voice assistants
- Smart speakers
- Noise pollution monitoring
- Security systems
- Acoustic event detection

17 Inertial Sensors

Inertial sensors measure motion, acceleration, rotation, and orientation.

These sensors are extensively used in mobile devices, robotics, drones, autonomous systems, and wearable devices.

17.1 Accelerometers

An accelerometer measures acceleration along one or more axes.

17.1.1 Working Principle

Most modern accelerometers are MEMS-based devices.

A proof mass is suspended within the sensor structure. When acceleration occurs, the proof mass experiences displacement, causing measurable changes in capacitance.

According to Newton's Second Law,

$$[F = ma]$$

where

F Applied force

m Mass

a Acceleration

17.1.2 Types

- Single-axis accelerometers
- Two-axis accelerometers
- Three-axis accelerometers

17.1.3 Applications

- Activity monitoring
- Fall detection
- Smartphone orientation sensing
- Structural health monitoring

17.2 Gyroscopes

A gyroscope measures angular velocity.

17.2.1 Working Principle

MEMS gyroscopes exploit the Coriolis effect.

When a vibrating mass undergoes rotation, Coriolis forces generate measurable displacement that can be converted into electrical signals.

17.2.2 Characteristics

- Measures rotational motion
- High sensitivity
- Fast response
- Suitable for navigation applications

17.2.3 Applications

- Drone stabilization
- Robotics
- Navigation systems
- Virtual reality devices

17.3 Magnetometers

Magnetometers measure the strength and direction of magnetic fields.

17.3.1 Applications

- Electronic compasses
- Navigation systems
- Geophysical surveys
- Autonomous vehicles

17.4 Inertial Measurement Unit (IMU)

An Inertial Measurement Unit (IMU) combines multiple inertial sensors into a single package.

Typical IMUs contain:

- Accelerometer
- Gyroscope
- Magnetometer (optional)

17.4.1 Example: MPU6050

The MPU6050 integrates:

- Three-axis accelerometer
- Three-axis gyroscope

The device communicates using the I²C protocol.

17.4.2 Applications

- Drones
- Robotics
- Motion tracking
- Wearable devices
- Autonomous vehicles

18 Position and Location Sensors

Position sensing is a fundamental requirement in tracking and navigation applications.

18.1 Global Positioning System (GPS)

The Global Positioning System (GPS) is a satellite-based navigation system that provides location and timing information anywhere on Earth.

18.1.1 Working Principle

A GPS receiver receives signals from multiple satellites orbiting the Earth.

The receiver determines its position through trilateration using the measured distances from at least four satellites.

18.1.2 Parameters Measured

- Latitude
- Longitude
- Altitude
- Speed
- Time

18.1.3 Advantages

- Global coverage
- High positioning accuracy
- Continuous operation

18.1.4 Limitations

- Reduced accuracy indoors
- Signal blockage in urban canyons
- Higher power consumption

18.1.5 Applications

- Fleet management
- Vehicle tracking
- Asset monitoring
- Navigation systems
- Smart transportation

19 Comparative Analysis of Common IoT Sensors

Table 2: Comparison of Common IoT Sensors

Sensor	Parameter Measured	Output Type	Interface
LM35	Temperature	Analog	ADC
DS18B20	Temperature	Digital	One-Wire
DHT22	Temperature, Humidity	Digital	Single-Wire
LDR	Light Intensity	Analog	ADC
BH1750	Light Intensity	Digital	I ² C
PIR Sensor	Motion Detection	Digital	GPIO
HC-SR04	Distance	Digital Pulse	GPIO
BMP280	Pressure, Temperature	Digital	I ² C/SPI

Sensor	Parameter Measured	Output Type	Interface
MQ-135	Air Quality	Analog	ADC
MPU6050	Acceleration, Rotation	Digital	I ² C
GPS Module	Location	Digital	UART

20 What is an Actuator?

An actuator is a device that converts electrical, hydraulic, pneumatic, thermal, or chemical energy into a physical action or movement. It serves as the output element of a control system and executes commands generated by a controller.

Formally, an actuator may be defined as:

An actuator is a transducer that transforms control signals into physical actions capable of affecting the environment.

The control signal may originate from a microcontroller, embedded computer, programmable logic controller (PLC), or cloud-based control system.

20.1 Role of Actuators in IoT Systems

In IoT architectures, actuators perform actions based on decisions derived from sensor data and computational algorithms.

Typical examples include:

- Switching electrical appliances ON or OFF.
- Opening or closing valves in irrigation systems.
- Adjusting motor speed in industrial machines.
- Controlling robotic movement.
- Operating smart locks and security systems.
- Regulating heating and cooling equipment.

Figure 7 illustrates the interaction between sensors, controllers, and actuators.



Figure 7: Basic Sensor–Controller–Actuator Interaction

21 Characteristics of Actuators

Selection of an actuator depends upon several performance characteristics.

21.1 Force and Torque

Force represents the linear mechanical output generated by an actuator, whereas torque represents rotational output.

Examples:

- Linear actuators generate force.
- Electric motors generate torque.

21.2 Speed

Speed refers to the rate at which an actuator performs a movement.

Examples include:

- Rotational speed of a motor measured in revolutions per minute (RPM).
- Extension speed of a linear actuator measured in millimeters per second.

21.3 Accuracy

Accuracy indicates how closely the actuator reaches a desired position or operating point.

High-precision applications such as robotic surgery and industrial automation require highly accurate actuators.

21.4 Resolution

Resolution defines the smallest change in output that an actuator can produce.

Stepper motors, for example, provide discrete angular movements with well-defined resolution.

21.5 Power Consumption

Power consumption is particularly important in battery-operated IoT systems where energy efficiency directly affects operational lifetime.

21.6 Response Time

Response time is the interval between receiving a control signal and producing the intended action.

Fast response is essential in applications such as autonomous vehicles and industrial safety systems.

22 Classification of Actuators

Actuators may be classified according to the source of energy used to produce motion.

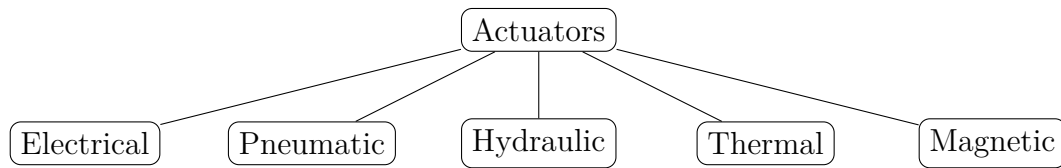


Figure 8: Major Categories of Actuators

The major actuator categories are discussed in the following sections.

23 Electrical Actuators

Electrical actuators are the most widely used actuators in IoT systems because they can be easily interfaced with microcontrollers and embedded platforms.

Advantages include:

- Easy electronic control.
- High efficiency.
- Compact size.
- Compatibility with digital systems.
- Low maintenance requirements.

Common electrical actuators include:

- DC Motors
- AC Motors
- Servo Motors
- Stepper Motors
- Solenoids
- Relays

24 DC Motors

Among all electrical actuators, Direct Current (DC) motors are the most commonly used devices for producing rotational motion. They are widely employed in IoT-based automation systems because they are inexpensive, easy to control, and readily available in a variety of power ratings.

Examples of applications include:

- Smart fans
- Robotic vehicles
- Conveyor systems
- Automated curtains
- Water pumps
- Smart door mechanisms

24.1 Construction of a DC Motor

A typical DC motor consists of the following components:

1. **Stator:** Produces a stationary magnetic field.
2. **Rotor (Armature):** Rotating part carrying conductors.
3. **Commutator:** Reverses current direction in the armature.
4. **Brushes:** Provide electrical contact between stationary and rotating parts.
5. **Shaft:** Transfers mechanical power to external loads.

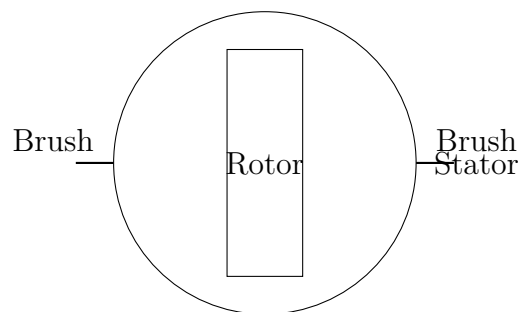


Figure 9: Simplified Structure of a DC Motor

24.2 Working Principle

The operation of a DC motor is based on the principle that a current-carrying conductor placed within a magnetic field experiences a mechanical force.

The magnitude of force is given by:

$$F = BIL \sin \theta$$

where

F Force (N)

B Magnetic flux density (T)

I Current through conductor (A)

L Length of conductor (m)

θ Angle between field and conductor

When current flows through the armature conductors, electromagnetic forces generate torque that causes the rotor to rotate.

24.3 Types of DC Motors

DC motors can be classified into several categories.

24.3.1 Brushed DC Motor

Uses brushes and a commutator for current switching.

Advantages:

- Low cost
- Simple control
- Easy availability

Disadvantages:

- Brush wear
- Electrical noise
- Maintenance requirements

24.3.2 Brushless DC Motor (BLDC)

Uses electronic commutation instead of mechanical brushes.

Advantages:

- Higher efficiency
- Longer lifetime

- Lower maintenance
- Reduced electrical noise

Applications include drones, electric vehicles, and industrial automation systems.

24.4 Motor Driver Circuits

Microcontrollers cannot directly power most DC motors because motors require significantly higher current than microcontroller GPIO pins can provide.

Therefore, motor driver circuits are used.

Common motor driver ICs include:

- L293D
- L298N
- TB6612FNG
- DRV8833

These drivers perform:

- Current amplification
- Direction control
- Electrical isolation
- Protection against back EMF

24.5 H-Bridge Motor Control

An H-Bridge is a circuit configuration that enables a DC motor to rotate in both clockwise and counterclockwise directions.

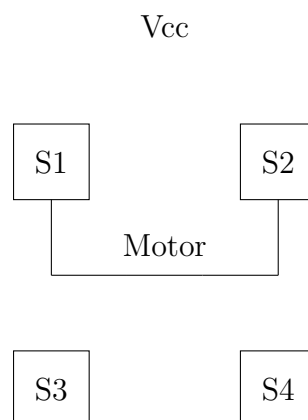


Figure 10: Conceptual H-Bridge Configuration

Operation:

- S1 and S4 ON → Clockwise rotation
- S2 and S3 ON → Counterclockwise rotation

Care must be taken to avoid short-circuit conditions.

24.6 Speed Control Using PWM

Pulse Width Modulation (PWM) is the most common technique used to control DC motor speed.

The average voltage applied to the motor is controlled by adjusting the duty cycle of a digital signal.

$$\text{Duty Cycle} = \frac{T_{ON}}{T} \times 100\%$$

where

T_{ON} ON duration

T Total period

Examples:

Duty Cycle	Effective Voltage (% of Supply)
25%	25%
50%	50%
75%	75%
100%	Full Voltage

Higher duty cycles generally produce higher motor speeds.

24.7 Back Electromotive Force (Back EMF)

As a DC motor rotates, it acts as a generator and produces a voltage opposing the applied voltage.

This induced voltage is called Back EMF.

According to Lenz's Law:

$$E_b \propto \omega$$

where

E_b Back EMF

ω Angular velocity

Back EMF helps regulate motor current and protects the motor from excessive power consumption.

24.8 IoT Applications of DC Motors

DC motors are extensively used in IoT systems requiring controlled movement.

Examples include:

- Smart robotic platforms
- Automated blinds and curtains
- Smart parking barriers
- Conveyor automation
- Home automation systems
- Agricultural irrigation mechanisms

25 Servo Motors

A servo motor is a specialized actuator designed to provide precise control over angular position, velocity, and acceleration. Unlike ordinary DC motors that rotate continuously, servo motors are capable of moving to a specific position and maintaining that position accurately.

Servo motors are widely used in:

- Robotics
- CNC machines
- Camera stabilization systems
- Smart locks
- Antenna positioning systems
- Industrial automation

Because of their precision and ease of control, servo motors are among the most important actuators used in IoT and embedded systems.

25.1 Construction of a Servo Motor

A servo motor is not merely a motor; it is a complete closed-loop control system packaged into a compact unit. A typical servo motor contains the following components:

1. DC Motor
2. Gear Reduction Mechanism

3. Position Sensor (usually a potentiometer)
4. Control Circuit
5. Output Shaft

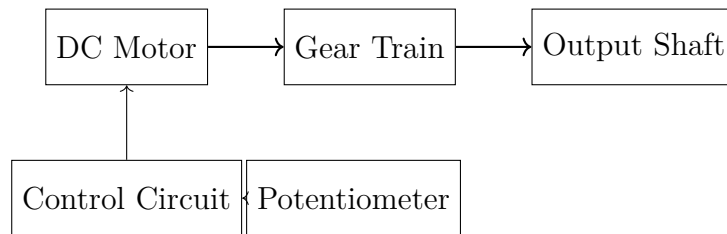


Figure 11: Simplified Internal Structure of a Servo Motor

The gear train reduces motor speed and increases torque. The potentiometer continuously measures the shaft position and provides feedback to the control circuit.

25.2 Working Principle

Servo motors operate using feedback control.

The control circuit continuously performs the following operations:

1. Receive desired position from the controller.
2. Measure current shaft position using the potentiometer.
3. Calculate the position error.
4. Drive the motor until the error becomes nearly zero.

The error signal is given by

$$e(t) = \theta_d - \theta_a$$

where

θ_d Desired angular position

θ_a Actual angular position

When the desired and actual positions become equal, the motor stops and holds its position.

25.3 Servo Control Using PWM

Most hobby servos are controlled using Pulse Width Modulation (PWM).

Unlike DC motor speed control, PWM in servos is used to specify position rather than speed.

Typically:

Pulse Width	Shaft Position
1.0 ms	0°
1.5 ms	90°
2.0 ms	180°

The PWM signal is usually repeated every 20 ms.

$$f = \frac{1}{20 \times 10^{-3}}$$

$$f = 50 \text{ Hz}$$

Thus, most servo motors operate using a PWM control frequency of approximately 50 Hz.

25.4 Servo Positioning Example

Suppose an IoT-based smart ventilation system requires a window flap to open halfway.

The controller may generate a PWM signal corresponding to

$$\theta = 90^\circ$$

The servo rotates until the shaft reaches the desired angle and then maintains that position using internal feedback.

25.5 Advantages of Servo Motors

- High positional accuracy
- Built-in feedback mechanism
- Easy control using microcontrollers
- High torque-to-size ratio
- Fast response

25.6 Limitations of Servo Motors

- Limited rotational range
- More expensive than ordinary DC motors
- Increased complexity
- Continuous holding consumes power

25.7 Popular Servo Motors

Some widely used servo motors in educational and IoT projects include:

Model	Typical Rotation Range
SG90	0°–180°
MG90S	0°–180°
MG996R	0°–180°
DS3218	High Torque Servo

The SG90 servo is one of the most popular actuators used in embedded system laboratories because of its low cost and ease of interfacing.

25.8 IoT Applications of Servo Motors

Servo motors are commonly used in:

- Smart door locks
- Camera pan-tilt systems
- Robotic arms
- Solar panel tracking systems
- Automated valves
- Smart agriculture equipment

26 Stepper Motors

A stepper motor is an electromechanical actuator that converts electrical pulses into discrete angular movements called *steps*.

Unlike ordinary motors that rotate continuously, stepper motors move in precise increments. This makes them suitable for applications requiring accurate positioning.

Examples include:

- 3D printers
- CNC machines
- Industrial automation systems
- Camera positioning systems
- Robotic platforms
- Smart manufacturing equipment

26.1 Basic Principle of Operation

A stepper motor rotates because its stator windings are energized in a specific sequence.

Each input pulse causes the rotor to move by a fixed angle known as the *step angle*.

$$\text{Angular Position} = N \times \theta_s$$

where

N Number of input pulses

θ_s Step angle

Thus, motor position can be controlled without requiring continuous feedback.

26.2 Construction of a Stepper Motor

A typical stepper motor consists of:

1. Stator windings
2. Rotor
3. Shaft
4. Bearings
5. Housing

The stator contains multiple electromagnetic coils arranged around the rotor.

The rotor may be:

- Permanent magnet
- Soft iron
- Hybrid structure

26.3 Types of Stepper Motors

Stepper motors are generally classified into three categories.

26.3.1 Permanent Magnet Stepper Motor

Uses a permanent magnet rotor.

Characteristics:

- Moderate accuracy
- Good torque
- Low cost

26.3.2 Variable Reluctance Stepper Motor

Uses a soft iron rotor.

Characteristics:

- High speed operation
- Lower torque
- Simple construction

26.3.3 Hybrid Stepper Motor

Combines features of permanent magnet and variable reluctance motors.

Characteristics:

- High precision
- High torque
- Excellent performance

Most industrial stepper motors belong to this category.

26.4 Step Angle

The step angle determines the angular movement produced by one input pulse.

$$\theta_s = \frac{360^\circ}{N_s}$$

where

N_s Number of steps per revolution

Example:

If a motor requires 200 steps per revolution,

$$\theta_s = \frac{360^\circ}{200} = 1.8^\circ$$

Therefore, every pulse rotates the shaft by 1.8° .

26.5 Stepping Modes

Stepper motors can operate in several stepping modes.

26.5.1 Full-Step Mode

One phase or two phases are energized at a time.

Characteristics:

- Maximum torque
- Simpler control
- Lower resolution

26.5.2 Half-Step Mode

Alternates between single-phase and dual-phase excitation.

Characteristics:

- Double resolution
- Smoother motion
- Improved positioning accuracy

26.5.3 Microstepping

Current in windings is varied continuously.

Characteristics:

- Very smooth motion
- High precision
- Reduced vibration

Microstepping is widely used in CNC and robotic applications.

26.6 Unipolar and Bipolar Stepper Motors

Stepper motors are commonly classified according to the winding arrangement used for excitation.

26.6.1 Unipolar Stepper Motor

A unipolar stepper motor contains center-tapped windings. Current flows through only one half of a winding at a time.

Characteristics:

- Simpler drive circuitry
- Easier interfacing with microcontrollers

- Lower torque compared to bipolar motors
- Popular in educational projects

A common example is the **28BYJ-48** stepper motor.

26.6.2 Bipolar Stepper Motor

A bipolar stepper motor does not use center taps. Current must be reversed through the winding to change magnetic polarity.

Characteristics:

- Higher torque
- Better efficiency
- More complex drive circuitry
- Widely used in industrial systems

Because current reversal is required, bipolar motors typically use H-bridge driver circuits.

26.7 Stepper Motor Excitation Sequence

The stator windings must be energized in a specific sequence to produce rotation.

Consider a four-phase stepper motor with windings A, B, C, and D.

26.7.1 Full-Step Sequence

Step	A	B	C	D
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1

Repeating this sequence causes continuous rotation.

26.7.2 Half-Step Sequence

Step	A	B	C	D
1	1	0	0	0
2	1	1	0	0
3	0	1	0	0
4	0	1	1	0
5	0	0	1	0
6	0	0	1	1
7	0	0	0	1
8	1	0	0	1

Half-step operation provides smoother motion and twice the angular resolution.

26.8 Stepper Motor Drivers

Since stepper motors require significant current and precise excitation sequences, dedicated driver circuits are used.

Functions of a stepper driver include:

- Current amplification
- Phase sequencing
- Current limiting
- Direction control
- Speed control

Common driver modules include:

- ULN2003
- A4988
- DRV8825
- TMC2208
- TMC2209

26.9 ULN2003 Driver

The ULN2003 is a Darlington transistor array commonly used with the 28BYJ-48 stepper motor.

Features:

- Seven transistor driver stages
- Built-in flyback diodes
- Simple interfacing
- Compatible with Arduino and ESP32 boards

The ULN2003 acts as a current amplifier, allowing the microcontroller to control the stepper motor safely.

26.10 A4988 Driver

The A4988 is one of the most widely used drivers for bipolar stepper motors.

Major features include:

- Full-step operation
- Half-step operation
- Microstepping support
- Adjustable current limiting
- Direction control
- Over-temperature protection

Only two control signals are typically required:

- STEP
- DIRECTION

Each pulse on the STEP pin advances the motor by one step.

26.11 Advantages of Stepper Motors

- Precise position control
- High repeatability
- Good low-speed torque
- Digital control compatibility
- Suitable for open-loop operation

26.12 Limitations of Stepper Motors

- Lower efficiency
- Resonance and vibration
- Torque decreases at high speed
- Continuous current consumption

26.13 IoT Applications of Stepper Motors

Stepper motors are commonly found in:

- Automated manufacturing systems
- Smart positioning platforms
- Precision agriculture equipment
- Camera control systems
- 3D printers
- Robotic arms

27 Relays

Many IoT systems need to control high-voltage or high-current devices such as lamps, pumps, fans, heaters, and industrial machinery.

Since microcontroller pins can typically supply only a few milliamperes of current at low voltage, a switching interface is required. Relays provide this capability.

27.1 Definition

A relay is an electrically operated switch that uses an electromagnetic mechanism to open or close electrical contacts.

Relays allow low-power electronic circuits to control high-power electrical loads safely.

27.2 Construction of an Electromagnetic Relay

A typical relay consists of:

1. Electromagnetic coil
2. Iron core
3. Armature
4. Spring mechanism
5. Electrical contacts

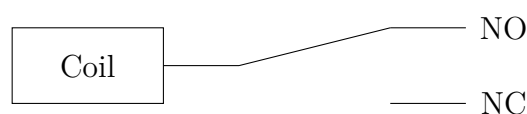


Figure 12: Conceptual Electromagnetic Relay Structure

27.3 Working Principle

When current flows through the coil, a magnetic field is generated.

The magnetic field attracts the armature, causing the switch contacts to change state.

When the coil current is removed:

- The magnetic field collapses.
- The spring returns the armature to its original position.
- Contacts return to their default state.

Thus, electrical isolation exists between the control circuit and the load circuit.

27.4 Relay Contact Terminology

The three most common relay terminals are:

- COM (Common)
- NO (Normally Open)
- NC (Normally Closed)

27.4.1 Normally Open (NO)

The circuit remains disconnected until the relay is energized.

Applications:

- Lamp control
- Motor switching
- Smart appliance control

27.4.2 Normally Closed (NC)

The circuit remains connected until the relay is energized.

Applications:

- Safety systems
- Alarm circuits
- Emergency shutdown systems

27.5 Relay Driver Circuit

A microcontroller cannot directly energize most relay coils because:

- Coil current exceeds GPIO capability.
- Back EMF can damage the controller.

Therefore a transistor driver circuit is normally used.

Typical components include:

- NPN transistor
- Base resistor
- Flyback diode
- Relay coil

The flyback diode protects the transistor and microcontroller from voltage spikes generated when the relay coil is switched OFF.

27.6 Solid State Relay (SSR)

A Solid State Relay performs switching electronically without moving parts.

Advantages:

- Fast switching
- Long lifetime
- Silent operation
- No mechanical wear

Disadvantages:

- Higher cost
- Leakage current
- Heat generation

SSRs are commonly used in industrial automation and smart energy systems.

27.7 Opto-Isolated Relay Modules

Many commercially available relay modules include opto-isolation to provide additional protection between the microcontroller and the load circuit.

An optocoupler (or opto-isolator) transfers signals using light rather than direct electrical connection.

A typical opto-isolated relay module contains:

- Input LED
- Phototransistor
- Driver transistor
- Relay
- Flyback diode

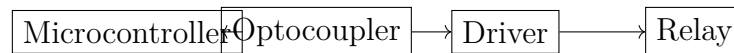


Figure 13: Simplified Opto-Isolated Relay Interface

Advantages of opto-isolation include:

- Protection against voltage spikes
- Improved electrical isolation
- Reduced electromagnetic interference
- Increased system reliability

Opto-isolated relay modules are widely used in industrial IoT deployments where electrical noise is significant.

27.8 IoT Applications of Relays

Relays remain one of the most important actuators in IoT systems because they allow digital electronics to control high-power equipment.

Examples include:

- Smart home appliance control
- Intelligent lighting systems
- Water pump automation
- Industrial machine control
- Smart energy management
- Building automation systems

28 Solenoids

A solenoid is an electromagnetic actuator that converts electrical energy into linear mechanical motion.

Unlike motors, which generally produce rotational movement, solenoids produce a pushing or pulling action.

Common applications include:

- Electronic door locks
- Valve control systems
- Vending machines
- Industrial automation
- Smart access systems

28.1 Construction of a Solenoid

A typical solenoid consists of:

1. Cylindrical coil
2. Ferromagnetic plunger
3. Return spring
4. Housing

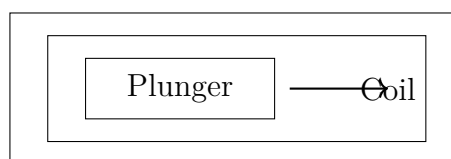


Figure 14: Simplified Solenoid Structure

28.2 Working Principle

When current flows through the coil:

1. A magnetic field is generated.
2. The plunger experiences a magnetic force.
3. The plunger moves toward the center of the coil.
4. Mechanical work is performed.

When power is removed:

- The magnetic field collapses.
- The spring returns the plunger to its initial position.

This motion can be used to operate locks, switches, and valves.

28.3 Types of Solenoids

28.3.1 Pull-Type Solenoid

The plunger is pulled inward when energized.

Applications:

- Door locks
- Mechanical latches
- Switching systems

28.3.2 Push-Type Solenoid

The plunger extends outward when energized.

Applications:

- Actuation mechanisms
- Industrial machinery
- Control systems

28.4 Advantages of Solenoids

- Simple construction
- Fast operation
- Compact size
- Easy digital control

28.5 Limitations of Solenoids

- Limited travel distance
- High current requirement
- Heat generation during continuous operation

28.6 IoT Applications of Solenoids

Examples include:

- Smart locks
- Automated dispensing systems
- Smart irrigation valves
- Industrial process automation
- Access control systems

29 Linear Actuators

A linear actuator is a device that produces controlled linear motion instead of rotational motion.

Linear actuators are extensively used when an object must move along a straight path.

Applications include:

- Adjustable hospital beds
- Smart furniture
- Solar tracking systems
- Industrial automation
- Robotic positioning systems

29.1 Types of Linear Actuators

Based on their operating mechanism, linear actuators can be classified as:

1. Electric linear actuators
2. Hydraulic linear actuators
3. Pneumatic linear actuators

29.2 Electric Linear Actuator

Electric linear actuators usually consist of:

- Electric motor
- Gearbox

- Lead screw
- Sliding shaft

Rotational motion of the motor is converted into linear displacement through the lead screw mechanism.

29.3 Hydraulic Linear Actuator

Hydraulic actuators use pressurized fluid to generate force.

Characteristics:

- Very high force capability
- Suitable for heavy machinery
- Precise control under large loads

Applications include:

- Construction equipment
- Industrial presses
- Heavy automation systems

29.4 Pneumatic Linear Actuator

Pneumatic actuators use compressed air to produce motion.

Characteristics:

- Fast response
- Clean operation
- Simple construction

Applications include:

- Packaging systems
- Manufacturing equipment
- Material handling systems

29.5 Comparison of Common Actuators

Actuator	Motion Type	Precision	Typical Applications
DC Motor	Rotational	Moderate	Fans, robots, pumps
Servo Motor	Angular Positioning	High	Robotics, smart locks
Stepper Motor	Discrete Rotation	Very High	CNC, 3D printers
Relay	Switching	Binary	Appliance control
Solenoid	Linear Motion	Moderate	Valves, locks
Linear Actuator	Linear Motion	High	Automation systems

30 Introduction to Control Systems

Sensors and actuators alone cannot create intelligent systems. A mechanism is required to determine how actuator actions should respond to sensor measurements.

This mechanism is provided by a **control system**.

A control system is an arrangement of components that manages, commands, directs, or regulates the behavior of a device or process to achieve a desired objective.

Examples include:

- Maintaining room temperature using an air conditioner
- Controlling motor speed
- Regulating water level in a tank
- Maintaining vehicle speed using cruise control
- Automatic irrigation systems

30.1 Basic Elements of a Control System

A typical control system contains:

1. Reference Input (Setpoint)
2. Controller
3. Actuator

4. Process (Plant)
5. Sensor
6. Feedback Path



Figure 15: Basic Control System Structure

The objective of the control system is to ensure that the plant output follows the desired setpoint as closely as possible.

31 Open-Loop Control Systems

An open-loop control system performs actions without using feedback from the output.

The controller does not verify whether the desired result has actually been achieved.

Examples include:

- Fixed-time traffic signals
- Electric toaster
- Traditional washing machine timer
- Fixed-duration irrigation system

31.1 Block Diagram of an Open-Loop Control System

The general structure of an open-loop control system is shown in Figure 16.



Figure 16: Open-Loop Control System

In this configuration, the controller generates commands solely based on the input signal. The output is not measured or compared with the desired value.

31.2 Advantages of Open-Loop Systems

- Simple design
- Low implementation cost
- Easy maintenance
- Fast operation
- No sensor requirement

31.3 Disadvantages of Open-Loop Systems

- Low accuracy
- Unable to compensate for disturbances
- Sensitive to environmental changes
- Poor adaptability
- No automatic error correction

31.4 Example: Open-Loop Irrigation System

Consider an irrigation system programmed to operate a water pump for 10 minutes every morning.

The controller activates the pump according to time alone.

Possible issues include:

- Soil may already be wet.
- Rainfall may have occurred.
- Water requirements may vary.

Since no soil moisture feedback is used, the system cannot adjust its behavior.

32 Closed-Loop Control Systems

A closed-loop control system continuously monitors its output and adjusts its actions accordingly.

The output is measured using sensors and fed back to the controller.

The controller compares:

- Desired output (setpoint)
- Actual output (feedback)

The resulting error is used to modify actuator behavior.

Closed-loop systems form the foundation of modern automation and IoT applications.

32.1 Block Diagram of a Closed-Loop Control System

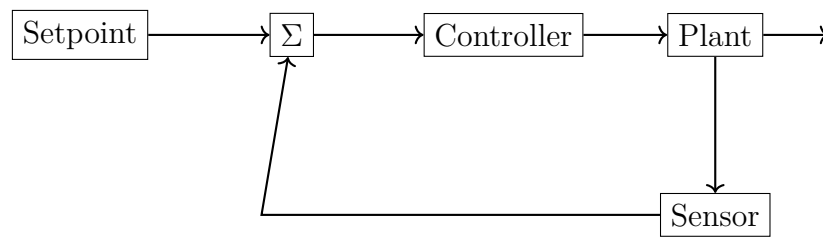


Figure 17: Closed-Loop Control System

The feedback path continuously informs the controller about the current system state.

32.2 Error Signal

The most important quantity in a feedback control system is the error signal.

$$e(t) = r(t) - y(t)$$

where

$r(t)$ Desired output (reference input)

$y(t)$ Actual output

$e(t)$ Error signal

The controller attempts to minimize this error.

If

$$e(t) = 0$$

then the system output exactly matches the desired value.

32.3 Advantages of Closed-Loop Systems

- High accuracy
- Automatic error correction
- Improved disturbance rejection
- Better stability under varying conditions
- Greater reliability

32.4 Disadvantages of Closed-Loop Systems

- Increased complexity
- Higher cost

- Additional sensors required
- More sophisticated design procedures

32.5 Example: Smart Thermostat

A smart thermostat measures room temperature continuously.

Suppose:

$$T_{set} = 25^{\circ}C$$

If the sensor reports:

$$T_{actual} = 29^{\circ}C$$

Then

$$e = 25 - 29 = -4^{\circ}C$$

The controller responds by increasing cooling action until the temperature approaches the setpoint.

32.6 Open-Loop versus Closed-Loop Systems

Parameter	Open-Loop	Closed-Loop
Feedback	Absent	Present
Accuracy	Low	High
Complexity	Low	Higher
Cost	Low	Higher
Error Correction	Not possible	Automatic
Disturbance Rejection	Poor	Good
Examples	Toaster, Timer	Thermostat, Cruise Control

33 Feedback in Control Systems

Feedback is the process of returning a portion of the output signal to the input for comparison and corrective action.

Feedback enables:

- Self-regulation
- Automatic correction

- Improved performance
- Increased accuracy

33.1 Types of Feedback

Feedback can be classified into two major categories.

33.1.1 Negative Feedback

Negative feedback subtracts the measured output from the reference input.

$$e(t) = r(t) - y(t)$$

Characteristics:

- Improves stability
- Reduces sensitivity to disturbances
- Improves accuracy
- Most widely used form of feedback

Examples include:

- Thermostats
- Motor speed control
- Industrial process control

33.1.2 Positive Feedback

Positive feedback adds the output signal to the input signal.

$$e(t) = r(t) + y(t)$$

Characteristics:

- Increases system gain
- May cause instability
- Useful in oscillators and regenerative circuits

Positive feedback is generally avoided in industrial control applications.

34 Transfer Function

The behavior of control systems is often described mathematically using transfer functions.

A transfer function represents the relationship between the output and input of a system in the Laplace domain.

$$G(s) = \frac{Y(s)}{U(s)}$$

where

$G(s)$ Transfer function

$Y(s)$ Output signal

$U(s)$ Input signal

Transfer functions simplify the analysis of dynamic systems.

34.1 Example

Consider a first-order system represented by

$$\tau \frac{dy(t)}{dt} + y(t) = Ku(t)$$

Taking the Laplace transform,

$$(\tau s + 1)Y(s) = KU(s)$$

Therefore,

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1}$$

where

K System gain

τ Time constant

Such first-order models frequently represent thermal systems, fluid systems, and simple motor dynamics.

35 Stability of Control Systems

A control system is said to be stable if its output remains bounded for a bounded input.

In practical terms, a stable system eventually settles to the desired operating condition after a disturbance.

35.1 Stable System

A stable system exhibits:

- Predictable behavior
- Finite output
- Convergence toward equilibrium

Example:

A thermostat gradually stabilizing room temperature at the desired value.

35.2 Unstable System

An unstable system exhibits:

- Increasing oscillations
- Diverging output
- Loss of control

Examples include:

- Poorly tuned drones
- Improper motor control loops
- Unstable robotic systems

35.3 Factors Affecting Stability

Several factors influence control system stability:

- Feedback gain
- Time delay
- Sensor noise
- Actuator saturation
- Controller parameters

Designing stable controllers is one of the primary objectives of control engineering.

36 PID Control

Among all industrial control techniques, the Proportional–Integral–Derivative (PID) controller is the most widely used.

PID controllers are employed in:

- Industrial automation
- Robotics
- Temperature control
- Motor speed regulation
- Autonomous vehicles
- Smart manufacturing

A PID controller combines three separate control actions:

1. Proportional (P)
2. Integral (I)
3. Derivative (D)

Each component contributes a different corrective action based on the error signal.

36.1 Proportional (P) Control

The proportional controller generates an output that is directly proportional to the current error.

$$u(t) = K_p e(t)$$

where

$u(t)$ Controller output

$e(t)$ Error signal

K_p Proportional gain

The proportional controller increases the control effort as the error increases.

36.1.1 Effect of Proportional Gain

Increasing K_p generally produces:

- Faster response
- Reduced steady-state error

- Increased sensitivity

However, excessively large values of K_p may cause:

- Oscillations
- Overshoot
- Instability

36.1.2 Example

Consider a motor speed control system.

Desired speed:

$$N_d = 1000 \text{ RPM}$$

Actual speed:

$$N_a = 900 \text{ RPM}$$

Therefore,

$$e = 100$$

If

$$K_p = 0.5$$

then

$$u = 0.5 \times 100 = 50$$

The controller increases motor drive effort proportionally to the error.

36.2 Integral (I) Control

The integral controller considers the accumulated error over time.

The control law is

$$u(t) = K_i \int e(t) dt$$

where

K_i Integral gain

The integral term continuously accumulates past errors.

36.2.1 Purpose of Integral Action

Integral control is primarily used to eliminate steady-state error.

Consider a temperature controller:

- Desired temperature: 25°C
- Actual temperature stabilizes at 24°C

The remaining 1°C error may persist indefinitely with proportional control alone.

Integral action accumulates this error and gradually drives it toward zero.

36.2.2 Advantages

- Eliminates steady-state error
- Improves long-term accuracy
- Compensates for constant disturbances

36.2.3 Disadvantages

- Slower response
- Increased overshoot
- Possibility of oscillations

36.3 Derivative (D) Control

The derivative controller predicts future behavior by considering the rate of change of error.

$$u(t) = K_d \frac{de(t)}{dt}$$

where

K_d Derivative gain

Derivative action responds to how rapidly the error is changing.

36.3.1 Purpose of Derivative Action

The derivative component acts as a damping mechanism.

If the error changes rapidly, derivative control applies corrective action before excessive deviation occurs.

This helps:

- Reduce overshoot
- Improve stability
- Increase damping

36.3.2 Advantages

- Improved transient response
- Reduced oscillations
- Better stability

36.3.3 Disadvantages

- Sensitive to noise
- Amplifies high-frequency sensor fluctuations
- Rarely used alone

37 Complete PID Controller

The PID controller combines proportional, integral, and derivative actions.

The controller output is

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

This equation forms the foundation of modern industrial control systems.

37.1 Block Diagram of a PID Controller

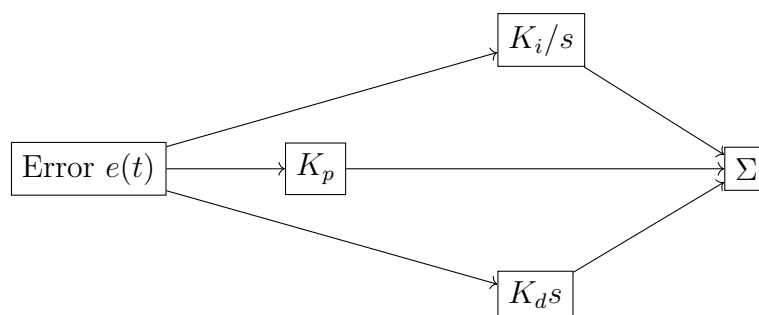


Figure 18: Conceptual PID Controller Structure

Each branch contributes a different corrective action.

37.2 Interpretation of PID Components

Component	Responds To	Primary Effect
Proportional	Present Error	Fast correction
Integral	Accumulated Error	Eliminates steady-state error

Derivative	Rate of Error Change	Predictive damping
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37.3 Performance Characteristics of Control Systems

Several performance measures are used to evaluate controller quality.

37.3.1 Rise Time

Rise time is the time required for the response to move from a small percentage to a large percentage of its final value.

Typically:

$$10\% \rightarrow 90\%$$

of the final value.

Smaller rise times indicate faster response.

37.3.2 Overshoot

Overshoot occurs when the response exceeds the desired value before settling.

$$\text{Overshoot} = \frac{Y_{max} - Y_{final}}{Y_{final}} \times 100\%$$

Excessive overshoot may damage equipment in practical systems.

37.3.3 Settling Time

Settling time is the time required for the output to remain within a specified error band around the final value.

Typical error bands:

- $\pm 2\%$
- $\pm 5\%$

37.3.4 Steady-State Error

Steady-state error is the remaining error after transient effects disappear.

$$e_{ss} = \lim_{t \rightarrow \infty} e(t)$$

A well-tuned PID controller aims to minimize this quantity.

37.4 Effects of PID Parameters

Parameter	Rise Time	Overshoot	Steady-State Error
Increase K_p	Decrease	Increase	Decrease
Increase K_i	Decrease	Increase	Eliminate
Increase K_d	Minor Effect	Decrease	Little Effect

38 PID Tuning

Selecting suitable values for K_p , K_i , and K_d is known as PID tuning.

Improper tuning may result in:

- Oscillations
- Slow response
- Instability
- Excessive overshoot

38.1 Manual Tuning

The simplest approach is trial and error.

General procedure:

1. Set $K_i = 0$ and $K_d = 0$.
2. Increase K_p gradually.
3. Add K_i to remove steady-state error.
4. Add K_d to reduce overshoot.

38.2 Ziegler–Nichols Method

One of the most famous PID tuning methods is the Ziegler–Nichols approach.

Procedure:

1. Set $K_i = 0$, $K_d = 0$.

2. Increase K_p until sustained oscillations occur.
3. Record the ultimate gain K_u .
4. Record oscillation period P_u .
5. Calculate PID parameters using empirical rules.

Although modern software tools often perform automatic tuning, the Ziegler–Nichols method remains historically important.

39 PID Applications in IoT Systems

PID control is extensively used in smart systems because many physical processes require continuous adjustment.

Applications include:

- Temperature control
- Motor speed control
- Drone stabilization
- Autonomous vehicles
- Industrial automation
- Smart agriculture

39.1 Example: Smart Greenhouse

A smart greenhouse may use:

- Temperature sensors
- Humidity sensors
- Fans
- Heaters
- Ventilation actuators

The controller continuously measures environmental conditions and adjusts actuators using PID algorithms to maintain optimal plant growth conditions.

39.2 Example: Motor Speed Control

Suppose a conveyor motor must operate at

$$1500 \text{ RPM}$$

The controller continuously measures actual speed using an encoder.
The PID controller adjusts motor voltage or PWM duty cycle until

$$e(t) \rightarrow 0$$

ensuring consistent speed despite load variations.

40 Actuator Interfacing with Embedded Platforms

In IoT systems, actuators are typically controlled using embedded platforms such as:

- Arduino
- ESP32
- Raspberry Pi
- STM32
- PIC Microcontrollers

These platforms generate digital control signals that are amplified by driver circuits before reaching actuators.

41 PWM Generation Basics

Pulse Width Modulation (PWM) is one of the most important techniques used in embedded systems for actuator control.

PWM allows a digital device to generate an analog-like output by rapidly switching a signal between HIGH and LOW states.

Applications include:

- DC motor speed control
- Servo motor positioning
- LED brightness control
- Power regulation
- Heating element control

41.1 Principle of PWM

A PWM signal consists of a periodic waveform whose ON duration can be varied while maintaining a constant period.

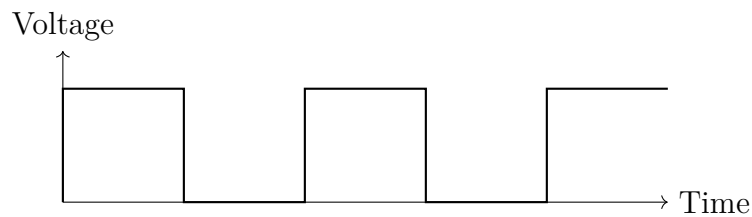


Figure 19: Example PWM Waveform

The duty cycle is defined as

$$D = \frac{T_{ON}}{T} \times 100\%$$

where

T_{ON} ON time

T Total period

41.2 Average Output Voltage

For a supply voltage V_s ,

$$V_{avg} = D \times V_s$$

where D is expressed as a fraction.

Example:

$$D = 0.6$$

$$V_s = 12V$$

$$V_{avg} = 0.6 \times 12$$

$$V_{avg} = 7.2V$$

Thus PWM effectively controls the average power delivered to a load.

41.3 PWM in Microcontrollers

Modern microcontrollers include dedicated timer peripherals capable of generating PWM signals.

Examples:

- Arduino Uno (ATmega328P)
- ESP32
- STM32
- PIC16F877A

PWM generation does not require continuous software intervention because hardware timers automatically produce the waveform.

42 Interfacing DC Motors with Arduino and ESP32

Since DC motors require higher current than microcontroller pins can supply, a motor driver circuit is required.

A typical arrangement consists of:

- Microcontroller
- Motor Driver
- External Power Supply
- DC Motor



Figure 20: DC Motor Interface Structure

42.1 Control Signals

Typical motor driver control signals include:

- Enable (PWM)
- Direction Input 1
- Direction Input 2

Direction control:

IN1	IN2	Direction
1	0	Clockwise
0	1	Counterclockwise
0	0	Stop
1	1	Brake

PWM is applied to the enable pin to control motor speed.

42.2 Typical Applications

- Mobile robots
- Smart fans
- Conveyor systems
- Automated gates

43 Interfacing Servo Motors with Arduino and ESP32

Servo motors require only three connections:

1. Power
2. Ground
3. PWM Control Signal

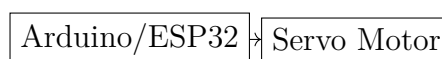


Figure 21: Servo Motor Interface

The control signal typically consists of:

- Frequency: 50 Hz
- Pulse width: 1–2 ms

Position control examples:

Pulse Width	Position
1 ms	0°
1.5 ms	90°
2 ms	180°

43.1 Important Considerations

Servo motors can draw significant current during startup.
Therefore:

- Use an external power supply.
- Avoid powering large servos directly from GPIO pins.
- Ensure common grounding.

44 Interfacing Stepper Motors with Arduino and ESP32

Stepper motors require sequential energization of multiple windings.

Consequently, dedicated driver circuits are necessary.

Common configurations include:

- 28BYJ-48 + ULN2003
- NEMA 17 + A4988

44.1 Interface Structure



Figure 22: Stepper Motor Interface

The microcontroller generates:

- Step pulses
- Direction signals

The driver converts these signals into appropriate winding currents.

44.2 Applications

- CNC machines
- Robotic arms
- Precision positioning systems
- 3D printers

45 Interfacing Relays with Arduino and ESP32

Relays are typically controlled through transistor driver stages.

The interface contains:

- GPIO output
- Driver transistor
- Relay coil
- Flyback diode

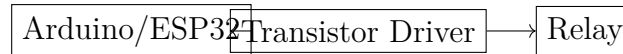


Figure 23: Relay Interface Structure

Relays enable low-power digital circuits to control:

- Lamps
- Pumps
- Fans
- Heaters
- Industrial equipment

46 Practical Design Considerations

Designing an actuator-based IoT system involves more than selecting sensors, controllers, and actuators. Practical implementation requires careful attention to electrical, thermal, electromagnetic, and safety-related issues.

Failure to consider these factors may lead to:

- Unreliable operation
- Reduced system lifetime
- Damage to electronic components
- Safety hazards
- Increased maintenance requirements

This section discusses important engineering considerations that should be addressed during the design of embedded and IoT systems.

46.1 Power Supply Considerations

One of the most common causes of actuator system failure is an inadequate power supply.

Actuators often require significantly more current than microcontrollers and sensors.

For example:

Device	Typical Current Requirement
ESP32	80–250 mA
Arduino Uno	30–70 mA
Small Servo Motor	200–800 mA
DC Motor	500 mA–Several A
Relay Coil	50–150 mA
Stepper Motor	300 mA–3 A

As a result, actuators are usually powered from dedicated power supplies rather than directly from microcontroller boards.

46.1.1 Voltage Rating

Every actuator must operate within its specified voltage range.

Examples:

- SG90 Servo: 4.8–6 V
- 28BYJ-48 Stepper Motor: 5 V
- Typical Relay Module: 5 V
- Industrial Solenoid: 12 V or 24 V

Operating below the rated voltage may result in:

- Reduced torque
- Sluggish response
- Failure to operate

Operating above the rated voltage may cause:

- Overheating
- Insulation breakdown
- Permanent damage

46.1.2 Current Capacity

Power supplies must be capable of delivering the maximum expected current.

For example, if three servo motors require

$$700mA$$

each during startup,

$$I_{total} = 3 \times 700mA = 2.1A$$

The power supply should be rated above this value with an appropriate safety margin.

46.1.3 Inrush Current

Many actuators draw a large current when starting.

This startup current is known as *inrush current*.

Examples include:

- DC motors
- Stepper motors
- Solenoids
- Relays

Power supplies must be capable of handling these temporary current surges.

46.1.4 Power Distribution

Good design practices include:

- Separate actuator and logic power rails.
- Use common grounding.
- Employ adequate conductor thickness.
- Minimize voltage drops in wiring.

A common architecture is shown below.

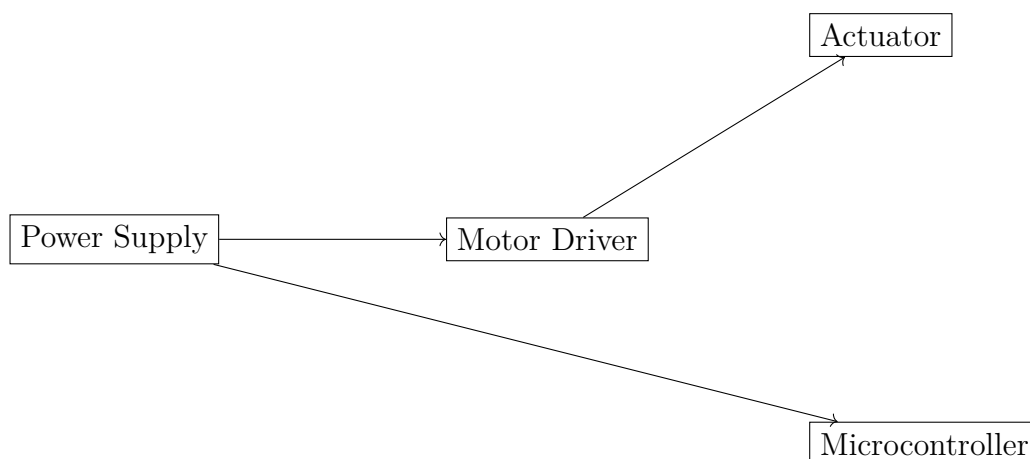


Figure 24: Separate Logic and Actuator Power Distribution

46.2 Flyback Diodes and Inductive Protection

Many actuators contain inductive components.

Examples include:

- Relay coils
- Solenoids
- DC motors
- Stepper motors

When current through an inductor is suddenly interrupted, a large voltage spike can be generated.

This phenomenon is explained by

$$V = L \frac{dI}{dt}$$

where

V Induced voltage

L Inductance

dI/dt Rate of current change

A rapid decrease in current may produce a very large voltage.

Such voltage spikes can:

- Damage transistors
- Reset microcontrollers
- Destroy driver circuits
- Generate electromagnetic interference

46.2.1 Flyback Diode

A flyback diode is connected across an inductive load in reverse bias during normal operation.

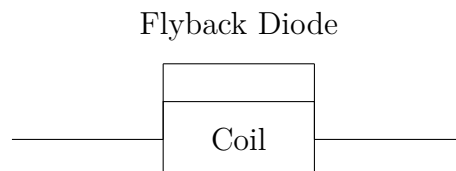


Figure 25: Conceptual Flyback Diode Across an Inductive Load

When the coil is switched OFF:

- The induced current circulates through the diode.
- Voltage spikes are suppressed.
- Driver circuits remain protected.

Common flyback diodes include:

- 1N4148
- 1N4007
- Schottky diodes

46.2.2 Additional Protection Methods

Other protection techniques include:

- RC snubber circuits
- TVS diodes
- MOV surge suppressors
- Driver ICs with integrated protection

46.3 Electromagnetic Interference (EMI)

Electromagnetic Interference (EMI) refers to unwanted electromagnetic energy that disrupts electronic operation.

Actuators are significant sources of EMI because they involve:

- Switching currents
- Rapid current changes
- Magnetic fields
- High-power operation

Common EMI-generating devices include:

- DC motors
- Stepper motors
- Relays
- Solenoids
- Switching power supplies

46.3.1 Effects of EMI

EMI may cause:

- Sensor measurement errors
- Communication failures
- Processor resets
- Data corruption
- System instability

46.3.2 EMI Reduction Techniques

Several design strategies can minimize EMI.

Decoupling Capacitors Capacitors placed near integrated circuits help stabilize supply voltages.

Typical values:

- 100 nF ceramic capacitor
- 10 μ F electrolytic capacitor

Shielding Conductive enclosures reduce electromagnetic radiation.

Examples include:

- Metal enclosures
- Shielded cables
- Grounded cable screens

Proper Grounding A low-impedance grounding network reduces noise propagation.

Good practices include:

- Star grounding
- Ground planes
- Short return paths

Cable Routing Signal cables should be routed away from power cables.

Recommendations:

- Minimize loop areas
- Avoid parallel routing
- Use twisted-pair conductors

46.4 Electrical Isolation

Electrical isolation prevents direct electrical connection between different sections of a system.

Isolation improves:

- Safety
- Noise immunity
- Equipment protection

Isolation is particularly important when microcontrollers interact with:

- Mains-powered equipment
- Industrial machinery
- High-voltage systems

46.4.1 Optocouplers

Optocouplers transfer signals using light.

A typical optocoupler contains:

- LED transmitter
- Phototransistor receiver



Figure 26: Basic Optocoupler Concept

Advantages include:

- Galvanic isolation
- Improved noise immunity
- Increased safety

46.4.2 Isolation Transformers

Isolation transformers provide electrical separation between circuits while allowing energy transfer through magnetic coupling.

Applications include:

- Industrial control systems
- Power supplies
- Medical equipment

46.4.3 Isolated Communication Interfaces

Modern industrial systems often employ isolated communication links such as:

- Isolated RS-485
- Isolated CAN Bus
- Industrial Ethernet interfaces

These interfaces improve reliability in electrically noisy environments.

46.5 Safety Considerations

Safety is a critical aspect of actuator-based system design.

Improper design may cause:

- Electrical shock
- Fire hazards
- Equipment damage
- Mechanical injury

46.5.1 Electrical Safety

When controlling mains-powered equipment:

- Maintain proper insulation.
- Use protective enclosures.
- Follow local electrical standards.
- Use appropriately rated relays and switches.

46.5.2 Overcurrent Protection

Protective devices should be incorporated to prevent excessive current flow.

Examples include:

- Fuses
- Circuit breakers
- Electronic current limiters

46.5.3 Thermal Protection

Actuators may generate substantial heat during operation.

Examples include:

- Motors
- Solenoids
- Power transistors

Protection methods include:

- Heat sinks
- Cooling fans
- Thermal sensors
- Automatic shutdown mechanisms

46.5.4 Mechanical Safety

Moving actuators can present physical hazards.

Potential risks include:

- Pinching
- Crushing
- Collision
- Entanglement

Protective measures include:

- Mechanical guards
- Emergency stop switches
- Travel limit switches
- Safety interlocks

46.5.5 Fail-Safe Design

A fail-safe system transitions into a safe state when faults occur.

Examples include:

- Automatic motor shutdown during overload
- Valve closure upon power failure
- Emergency stop activation during controller malfunction

Fail-safe mechanisms are essential in industrial automation, transportation systems, healthcare equipment, and critical infrastructure.

The practical considerations discussed in this section are fundamental to the design of reliable, safe, and robust IoT systems. Proper attention to power management, protection circuits, electromagnetic compatibility, isolation, and safety engineering ensures that actuators operate efficiently while minimizing risk to users and equipment.